1) $3 \cdot 2 + -2 + 4 + -1 = 7$

$\text{ReLU}(7) = 7$

2) $784 \cdot 128 + 128 \times 64 + 64 \times 5 = 109184$

3) It is a continuous linear transformation and activation function introduces non-linearity so it isn't expressed as a single matrix

4) L2 regularization

Lower alpha / Learning rate

5) $[-1, 0, -1, -1]$

6) $\frac{64-3}{1} + 1 = 62 \times 62$

$\frac{62}{2} = 31 \times 31$

$\frac{31-3}{1} + 1 = 29 \times 29$

7) Each layer sees a small patch and detects local patterns and as it goes deeper, neurons receptive field grows bigger.

8) 4,096,000 vs 288

It matters because of parameter efficiency

9) Man and telescope have more semantic meaning than with so it will assign more importance on those words

10) $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ $\begin{matrix} 1/\sqrt{2} \\ 0 \\ -\frac{1}{\sqrt{2}} \end{matrix}$ $\rightarrow \begin{bmatrix} e^{\frac{1}{\sqrt{2}}} \\ 0 \\ e^{-\frac{1}{\sqrt{2}}} \end{bmatrix}$

11) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$

12) Bert — Task classification

GPT — Generation, Text completion

13) $0 + 0 + 0.8^2 \cdot 5 + 0 + 0.8^4 \cdot 10$
$= 7.296$

14) $P(S_{t+1} | S_t) = P(S_{t+1} | S_{t-1} \dots S_0)$
Holds: chess position and all legal moves
Not holds: Poker

15) Credit in chess is only awarded at the end so moves should be penalized in what way? Bellman's equation $v(s) = r(s) + \gamma \max v(s')$ solves this

16) a) 10% are random
b) -10 per step
c) Higher early lower late since the agent knows more

17) $8 + 0.9 \cdot 10 = 17$
$17 - 4 = 13$
$4 + 0.9 \cdot 13 = 7.4$

18)	$s_0 \to s_1$	0	$0 + 0.5[0] = 0$	0	0
	$s_1 \to s_2$	0	$0 + 0.5[0] = 0$	0	0
	$s_2 \to s_3$	20	$0 + 0.5[20] = 10$	0	10
	$s_0 \to s_1$	0	$0 + 0.5[0] = 0$	0	10
	$s_1 \to s_2$	0	$0 + 0.5(0 + 10 - 0] = 5$	5	10
	$s_2 \to s_3$	20	$10 + 0.5(20 - 10] = 15$	15	15

19) Off policy is used to collect experience and is diff from policy being learned. It is greedy.
Sarsa:

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma Q(s',a') - Q(s,a)]$$

Sarsa uses a' while Q learning greedy over all actions

20) board game, Connect, as standard fit is a known